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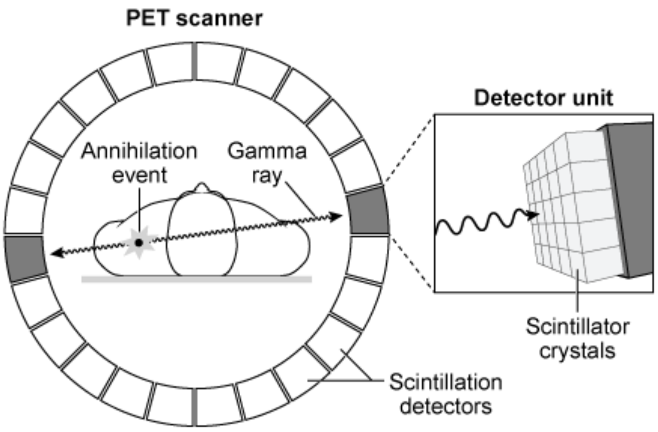

Figure 1 Basic setup and process of positron emission tomography

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Equation 1

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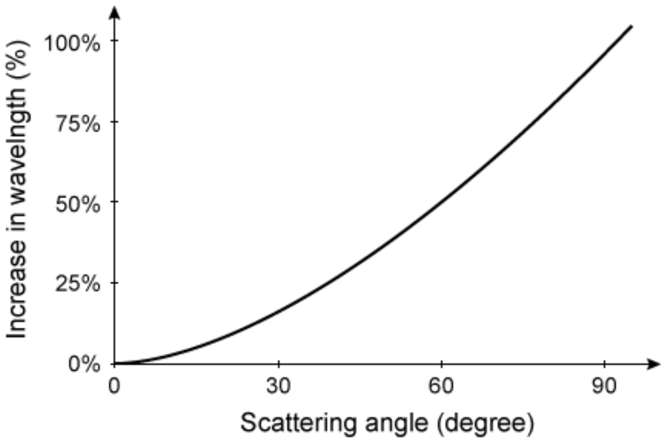


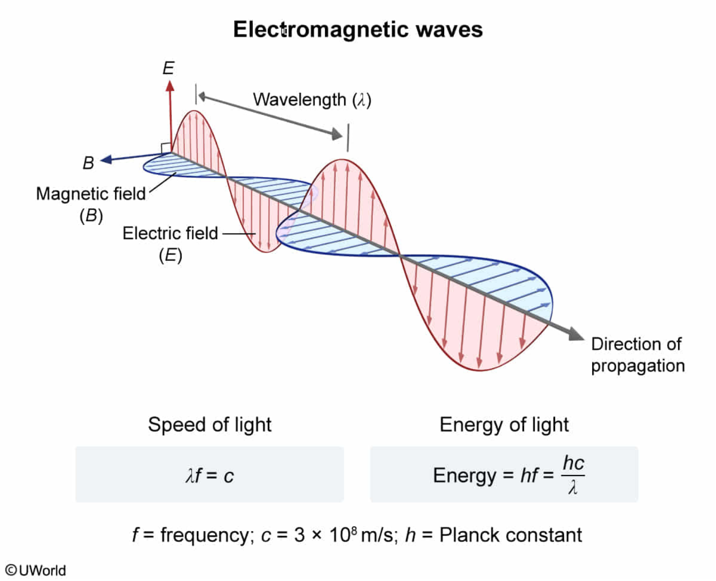
Figure 2 Percent change in photon wavelength due to Compton scattering

What is the frequency of the gamma rays emitted in PET? (Note: Use $h = 4 \times 10^{-15}$ eV·s.)

- A. 1.28×10^{20} Hz [65%]
- B. 1.28×10^{17} Hz [17%]
- C. 2.44×10^{-9} Hz [9%]
- D. 2.44×10^{-12} Hz [6%]

Correct
66% answered correctly

Explanation:



Electromagnetic radiation is propagated by oscillating particles called photons, which consist of oscillating electric and magnetic fields. Photons are packets of energy that travel at a constant speed of 3×10^8 m/s in a vacuum (known as the speed of light, *c*). The electromagnetic waves are transverse and propagate perpendicularly to each other. An electromagnetic wave is characterized by its wavelength and frequency ($\lambda f = c$), and the **energy** of a photon is directly **proportional** to its **frequency** *f* and the Planck constant *h*:

$$E = \mathrm{hf}$$

Rearranging the above equation to solve for frequency **(Choice D)**,

$$f = \frac{E}{h}$$

As stated in the passage, the gamma rays emitted in PET have 511 keV of energy. Using an energy *E* of 511 keV (5.11×10^5 eV) and the given value of *h*,

$$f = \frac{E}{h} = \frac{5.11 \times 10^5 \cancel{\mathrm{eV}}}{4 \times 10^{-15}\ \cancel{\mathrm{eV}} \cdot \mathrm{s}} \approx 1.28 \times 10^{20}\ \mathrm{s}^{-1}$$

This value is rewritten in SI units of frequency (1 Hz = 1 s^{-1}) as f = **1.28 × 10^{20} Hz**.

(Choice B) The prefix "k" in keV is the SI prefix *kilo*, which multiplies the value of the unit by 1,000. Therefore, 511 keV = 511,000 eV = 5.11 × 10^5 eV.

Educational objective:
Photons are oscillating packets of electromagnetic energy that are characterized by their wavelength and frequency. The frequency of a photon can be calculated from its energy: $f = \frac{E}{h}$.

Time spent:0 sec
Subject:
Physics

QID:400795
System:
Light and Sound

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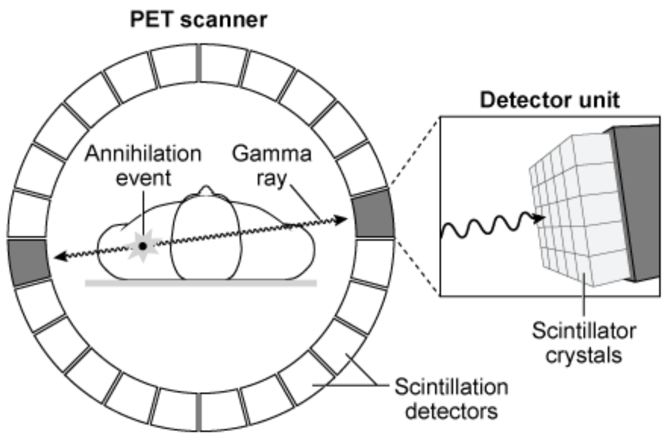

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Equation 1

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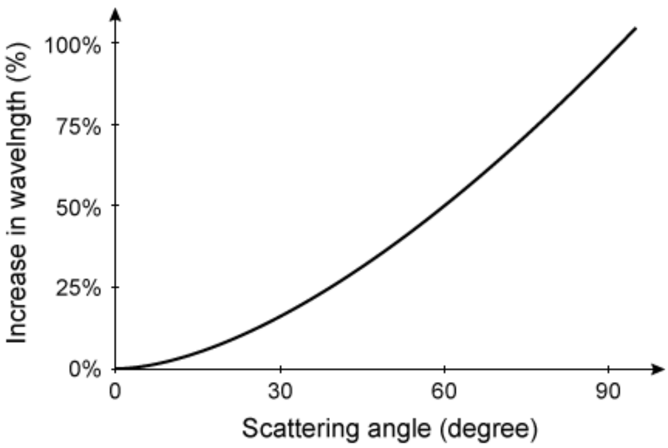


Figure 2 Percent change in photon wavelength due to Compton scattering

If the angle between a Compton-scattered photon and an electron is 60°, what is the energy of the scattered photon in terms of the original energy *E*?

A. $\frac{1}{2}E$ [52%]

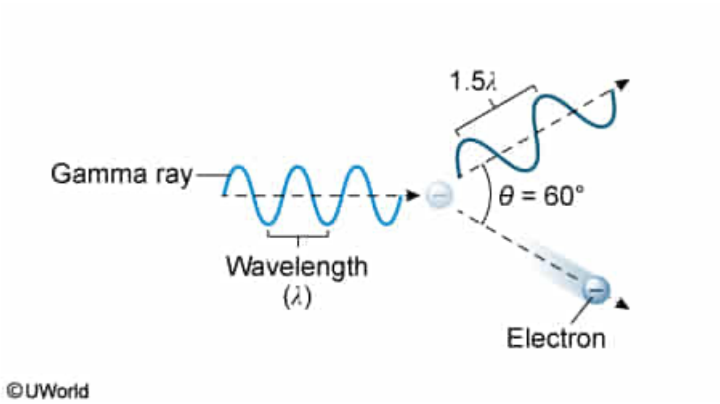
B. $\frac{2}{3}E$ [30%]

C. E [1%]

D. $\frac{3}{2}E$ [15%]

Correct
29% answered correctly

Explanation:



Photons are individual packets of energy that make up all forms of electromagnetic radiation. According to the passage, a **photon loses energy** to an electron when it undergoes Compton scattering rather than gaining or remaining constant in energy **(Choices C and D)**. The energy *E* of a photon is proportional to its frequency *f* and the Planck constant *h*:

$$E = \mathrm{hf}$$

Additionally, a photon's wavelength *λ* and frequency *f* are inversely related, and their product is equal to the speed of light ($c = 3.0 \times 10^8$ m/s):

$$\lambda f = c$$

Because the loss of photon energy is only given as a change in wavelength (Equation 1 and Figure 2), the above two equations are combined to eliminate frequency from the energy equation:

$$E = \frac{\mathrm{hc}}{\lambda}$$

Therefore, a photon's **energy and wavelength** are **inversely related**. From Figure 2, a scattering angle of 60° is observed when the photon

wavelength increases by 50%, which is equivalent to multiplying the wavelength by 1.5. Because photon energy is inversely related to wavelength, the photon **energy** must **decrease** by a **factor of 1.5**:

$$E_{\text{Scattered}} = \frac{\text{hc}}{1.5\ \lambda} = \frac{1}{1.5} \cdot \frac{\text{hc}}{\lambda} = \frac{1}{1.5} \cdot E$$

$$E_{\text{Scattered}} = \frac{E}{1.5} = \frac{E}{3/2} = \frac{2}{3}E$$

which is equivalent to **two thirds of the initial energy** *E* remaining.

(Choice A) Figure 2 gives the percent increase in wavelength. Therefore, the final wavelength is the initial wavelength λ plus the increase 0.5λ, which is 1.5λ.

Educational objective:
The energy of a photon is proportional to its frequency $\left(E = \text{hf}\right)$, and its frequency is inversely proportional to its wavelength $\left(\lambda f = c\right)$. Therefore, the change in energy of a photon is inversely proportional to its change in wavelength $\left(E = \frac{\text{hc}}{\lambda}\right)$.

Time spent:0 sec
Subject:
Physics

QID:400796
System:
Light and Sound

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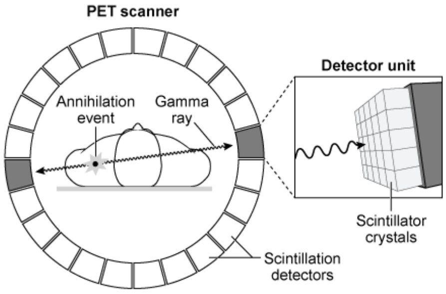

Figure 1 Basic setup and process of positron emission tomography

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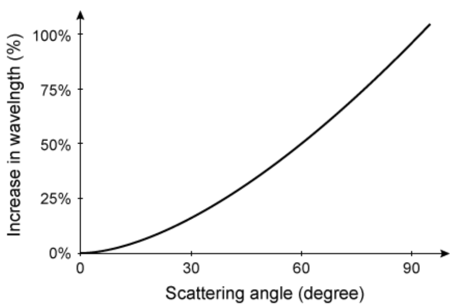


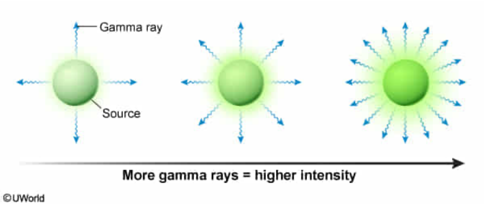
Figure 2 Percent change in photon wavelength due to Compton scattering

The radiopharmaceutical administered to a patient is carefully chosen to provide a specific number of positron emissions per second. If the rate of positron emissions is increased, which of the following can be expected?

- A. Higher individual gamma ray energy [15%]
- B. Higher gamma ray intensity [62%]
- C. Higher scintillation photon wavelength [8%]
- D. Higher scintillation photon velocity [14%]

Correct
61% answered correctly

Explanation:



A wave transfers energy through a medium in the form of oscillations. The intensity of light and sound waves is associated with their brightness or loudness, and is defined as the amount of power (energy per unit time) delivered per unit area:

$$\text{intensity} = \frac{\text{power}}{\text{area}} = \frac{\text{energy/time}}{\text{area}}$$

In other words, the **intensity** of electromagnetic radiation is **proportional to** the individual energy of each emitted particle and the **number of particles** emitted per unit time.

According to the passage, each positron produces two gamma rays when it annihilates with a nearby electron. **Increasing** the **rate of positron emissions** will increase the number of gamma rays produced per unit time, and therefore **increase the intensity** of gamma rays.

(Choice A) According to the passage, the annihilation events produce 511 keV gamma rays. Because only the number of positrons per unit time changes (not the process by which the gamma rays are generated), the *individual* energy of each emitted gamma ray remains the same.

(Choice C) According to the passage, the wavelength of the emitted light depends exclusively on the scintillation material used. Therefore, increasing the rate of positron emissions will not change the wavelength of the scintillation light.

(Choice D) The propagation speed of electromagnetic radiation depends only on the characteristics of the medium through which it travels. Because the medium did not change, the propagation speed of the scintillation photons will also remain unchanged.

Educational objective:
Intensity measures the amount of power (energy per unit time) delivered per unit area. The intensity of electromagnetic radiation increases with higher individual photon energy and higher emission rate.

Time spent:0 sec
Subject: Physics
QID:400797
System: Light and Sound

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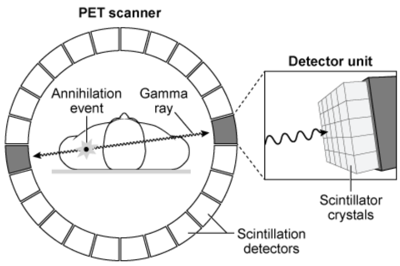

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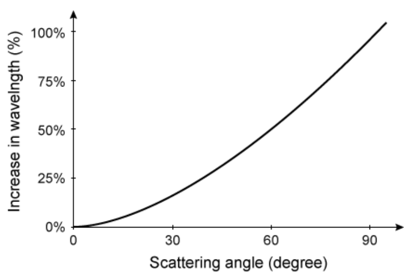

Figure 2 Percent change in photon wavelength due to Compton scattering

In which of the given scintillation materials will light travel the slowest?

A. Lu_2S_3:Ce [79%]

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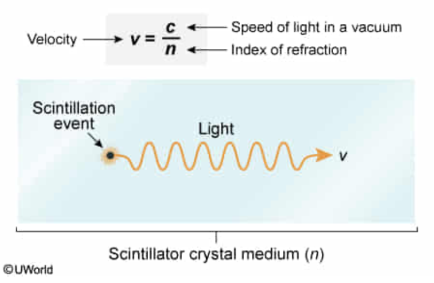
◯ B. BGO
◯ C. LSO [1%]
◯ D. BaF_2:Ce [18%]

Correct
80% answered correctly

Explanation:



The speed of light in a vacuum *c* is a constant, 3.0 × 10^8 m/s. However, when light travels through transparent media, the atoms in the material absorb and re-emit the particles of light (photons). Therefore, the velocity of light in transparent media is slower than in a vacuum. The ratio of the speed of light in a vacuum *c* to the speed of light in the material *v* is the material's index of refraction *n*:

$$n = \frac{c}{v}$$

The **velocity** of light through each crystal is **inversely proportional** to its **index of refraction**. Therefore, the scintillation **light** will be the **slowest** in the crystal with the **highest index of refraction**: **Lu_2S_3:Ce** (see Table 1).

Educational objective:
Light travels fastest in vacuum and is slower in transparent materials. The ratio of the speed of light in a vacuum *c* to the speed of light in a material *v* is the material's index of refraction $\left(n = \frac{c}{v}\right)$. Therefore, the higher the index of refraction, the slower light travels through a material.

Time spent:0 sec
Subject:
Physics

QID:400798
System:
Light and Sound

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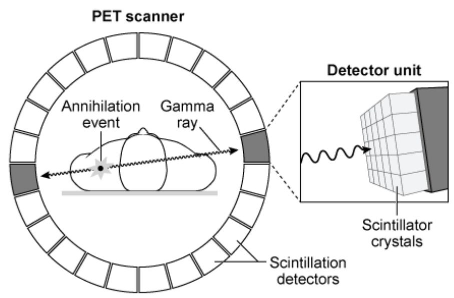

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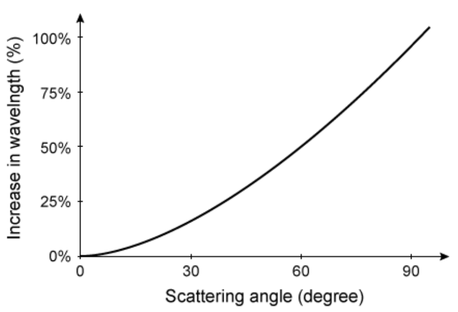

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Figure 2 Percent change in photon wavelength due to Compton scattering

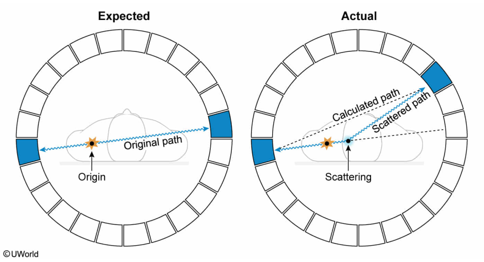
If one gamma photon out of a pair is Compton-scattered within the patient (not at the detector) and both gamma photons are still detected, which of the following would be seen?

I. The frequency of the scattered photon would be lower.
II. The point of origin would appear to be at a different location.
III. The velocity of the photon would decrease.

- A. II only [20%]
- B. III only [8%]
- ✔ C. I and II only [42%]
- D. I and III only [28%]

Correct
42% answered correctly

Explanation:



Electromagnetic radiation is propagated by oscillating particles of light called photons, and a photon's **energy** is directly **proportional** to its **frequency** ($E = hf$). As explained in the passage, gamma rays **lose energy** during Compton scattering because energy is transferred from the photon to the electron. Therefore, the scattered photon **frequency should be lower** than the incident photon frequency because its energy has decreased **(Number I)**.

As indicated in Figure 1, the localization of the positron source in PET relies on the assumption that the emitted gamma rays have trajectories 180° apart and therefore occur somewhere in the straight line between the two detectors. If Compton scattering occurs in the patient, the **trajectory** of the scattered gamma ray will be **altered**. As shown in the figure above, the calculated **point of origin** will be **different from the actual** location **(Number II)**.

(Number III) The propagation velocity of light is independent of the photon energy; it depends only on the medium through which it travels. Therefore, its velocity will not be affected by Compton scattering.

Educational objective:
The energy of a photon is proportional to its frequency: $E = hf$. Therefore, a photon's frequency will decrease as its energy decreases. The velocity of a photon's propagation is independent of the photon energy.

Time spent: 0 sec
Subject: Physics
QID: 400799
System: Light and Sound

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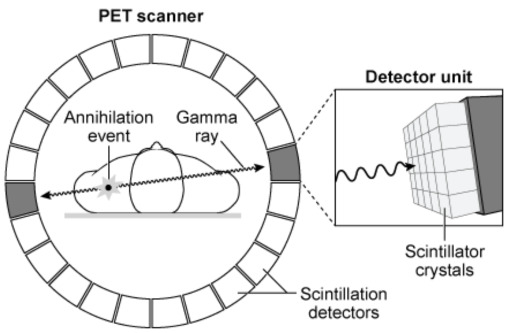

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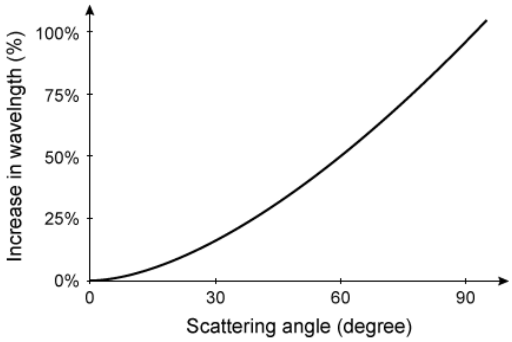


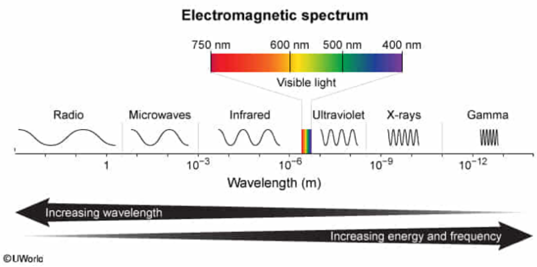
Figure 2 Percent change in photon wavelength due to Compton scattering

Which scintillation crystal emits light with a wavelength outside the visible spectrum?

- A. Lu_2S_3:Ce [3%]
- B. BGO
- C. LSO
- D. BaF_2:Ce [95%]

Correct
96% answered correctly

Explanation:



All electromagnetic radiation (including light) is made up of oscillating particles called photons, which are fully characterized by their wavelength and frequency. The electromagnetic spectrum is the entire range of electromagnetic radiation classified into different categories based on wavelength, frequency, or energy.

The **visible light spectrum** contains only the range of light that the human eye can detect. The wavelength of visible light ranges from about **400 to 750 nm** (1 nanometer = 10^{-9} m). From Table 1, BaF_2:Ce is the only crystal whose wavelength of 320 nm is outside the visible spectrum.

Educational objective:
The visible light spectrum is the part of the electromagnetic spectrum that the human eye can detect. Visible light has a wavelength between 400 and 750 nm.

Time spent:0 sec
Subject:
Physics

QID:400800
System:
Light and Sound